Mahmoud Souabni

Engineering Student in Industrial Computing & Automation – Final Year (Engineering Degree)

42 Avenue de la République
Mégrine 2033, Tunisia
+216 25 522 454
souabnimahmoud@gmail.com
linkedin.com/in/souabni-mahmoud-6621a8291

Profile

Final-year engineering student (M.Eng., Final Year) at INSAT, specializing in embedded systems, mobile robotics, and industrial automation. Completed a recent 2-month internship at CyberMech Systems configuring autonomous navigation (ROS 2, Nav2, SLAM) for a modular mobile robot. Additional experience in microcontrollers (STM32, ESP32, NXP), PCB design, and CAD. Seeking a final-year (capstone) internship for 2027 in robotics, embedded systems, or industrial automation.

Education

2024–2027 **Engineering Degree (Industrial Computing & Automation) – 3rd Year (in progress)**, *INSAT*, Tunis
Modules: Control Systems, Automation, C/C++, Microprocessor Architecture, Databases.

2022–2024 **Preparatory Cycle (Mathematics, Physics, Computer Science)**, *INSAT*, Tunis

Professional Experience

Summer 2026 (2 months) **Intern – Autonomous Navigation, AMR-X Robot**, *CyberMech Systems*, Tunis
Configured the ROS 2/Nav2 navigation stack for a modular mobile robot (AMR-X) in Gazebo: SLAM mapping, AMCL localization, path planning, obstacle avoidance, keepout zones, and mission execution via named stations. Diagnosed and corrected an odometry drift issue (kinematic calibration). Contributed to the CAD → URDF pipeline and to a market study of the AMR sector.

Summer 2025 **Observation Internship**, *Tunisair Technics*, Tunis
MRO: hands-on exposure to aircraft maintenance, avionics, and quality control.

Technical Skills

Programming	C/C++, Python, MATLAB, Java, SQL, Assembly
Robotics	ROS 2 (Jazzy) | Nav2, SLAM Toolbox, AMCL | Gazebo Harmonic, PyBullet | Git/colcon
Embedded	Arduino, ESP32, STM32, NXP | RTOS | Protocols (I2C, SPI, UART)
Automation	TIA Portal, Factory I/O, PL7 Pro, MATLAB/Simulink
Design	SolidWorks, URDF | EasyEDA, Proteus (PCB Design and Simulation)

Technical Projects

2025–2026 **Spider Robot – Final-Year Project** (INSAT) – Quadruped robot: 3D modeling, custom PCB, C firmware (servo control), and gait simulation in PyBullet.

2024 **AI-Based Beamforming** (MATLAB) – LSTM neural network for optimizing antenna beam direction.

Competitions & Robotics

2026 **NXP Cup Tunisia** – Programmed an autonomous car on an NXP microcontroller.

2025 **ECPP 2.0** (IEEE RAS INSAT) – Competitive programming contest focused on embedded systems.

2024–2025 **Line Follower Robot** – PCB and STM32/ESP32 firmware (Robolympix, Cyberbot, ENSI Robocup).

Activities & Leadership

2024–Pr. **Aerobotix INSAT** – Active member of the competitive robotics team.

2022–2024 **IEEE Student Branch** (CS/RAS) & **Securinets INSAT**.

Languages

Arabic: Native | **French:** Professional | **English:** Professional